HW9

November 27, 2025

Exercise 1 Prove the following form of Gronwall's inequality.

Let $\eta(t)$ be non-negative and absolutely continuous on $[0,T]$. Assume that the differential inequality

$$\eta'(t) \le \phi(t)\eta(t) + \psi(t), \quad \text{a.e. } t \in [0,T],$$

holds, where ϕ, ψ are non-negative and integrable on $[0,T]$. Show that

$$\eta(t) \le e^{\int_0^t \phi(s)\,ds}\big[\eta(0) + \int_0^t \psi(s)\,ds\big], \quad \forall t \in [0,T].$$

Note: η being absolutely continuous means that $\eta'(t)$ exists for a.e. $t \in [0,T]$ and

$$\eta(t) - \eta(0) = \int_0^t \eta(s)\,ds, \quad \forall t \in [0,T].$$

If you are not comfortable with this part of real analysis, you can do this exercise assuming $\eta \in \mathcal{C}^1$ and ignoring all the "a.e." above.

Exercise 2 Let Ω be a bounded domain and $u \in \mathcal{C}^{1,2}(\overline{\Omega_T})$ solve the equation

$$\begin{cases} \partial_t u(t,x) - \Delta u(t,x) + \sum_{i=1}^d b^i(t,x)\partial_{x_i} u(t,x) = f(t,x), & (t,x) \in \Omega_T, \\ u(t,x) = 0, & x \in \partial\Omega,\ t \ge 0, \\ u(0,x) = g(x), & x \in \bar{\Omega}. \end{cases}$$

Let $b^i(t,x), f \in \mathcal{C}(\overline{\Omega_T})$ and $g \in \mathcal{C}(\bar{\Omega})$.

1. For $u, v \in \mathcal{C}_0^2(\Omega)$, let

$$B[u,v;t] := \int_\Omega \nabla u \cdot \nabla v + \sum_{i=1}^d b^i \partial_{x_i} u \cdot v.$$

Show that

$$\big|B[u,v;t]\big| \le M\|u\|_{H_0^1}\|v\|_{H_0^1}$$

and

$$B[u,u;t] \ge \theta\|u\|_{H_0^1}^2 - M\|u\|_{L^2}^2$$

for some $M, \theta > 0$.

Hint: for the second one, note that

$$\|u\|_{L^2}\|u\|_{H_0^1} \le \frac{\varepsilon}{2}\|u\|_{H_0^1}^2 + \frac{1}{2\varepsilon}\|u\|_{L^2}^2$$

for every $\varepsilon > 0$ and choose ε properly.

2. Establish the energy estimate: for some C,

$$\max_{0\le t\le T}\|u(t,\cdot)\|_{L^2}+\|u\|_{L^2(0,T;H_0^1)}\le C\big[\|f\|_{L^2(0,T;L^2)}+\|g\|_{L^2}\big].$$

Hint: You should start with

$$\frac{1}{2}\frac{d}{dt}\int_\Omega u^2\,dx-\int_\Omega u\Delta u\,dx+\sum_{i=1}^d\int_\Omega b^i\partial_{x_i}u\,dx=\int_\Omega fu\,dx.$$

3. (optional) Also estimate the H^{-1} norm of u_t:

$$\|u_t\|_{L^2(0,T;H^{-1})}\le C\big[\|f\|_{L^2(0,T;L^2)}+\|g\|_{L^2}\big].$$

You can use the following definition: for $\varphi\in\mathcal{C}^2(\bar\Omega)$,

$$\|\varphi\|_{H^{-1}}=\sup_{v\in\mathcal{C}_0^\infty}\frac{\int_\Omega\varphi(x)v(x)\,dx}{\|\varphi\|_{H_0^1}}.$$

Ex1. 令 $\Phi(t) := \int_0^t \phi(s)\,ds$，$F(t) := \eta(t)e^{-\Phi(t)}$

由绝对连续性得，$F'(t) = \eta'(t)e^{-\Phi(t)} - \Phi(t)\eta(t)e^{-\Phi(t)}$，a.e. $t \in [0,T]$.

代入不等式得，$F'(t) \le \psi(t)e^{-\Phi(t)}$，a.e. $t \in [0,T]$.

积分，有 $F(t) - F(0) \le \int_0^t \psi(s)e^{-\Phi(s)}\,ds$，$\forall t \in [0,T]$.

即 $\eta(t)e^{-\Phi(t)} \le \eta(0) + \int_0^t \psi(s)\,ds$，$\forall t \in [0,T]$.

两边乘 $e^{\Phi(t)} > 0$ 即得结论.

Ex2. (1) 一方面，
$$\begin{aligned}|B[u,v;t]| &\le \int_\Omega \Big(|\nabla u||\nabla v| + \sum_{i=1}^d |b^i||\partial_{x_i}u||v|\Big)dx \\ &\le \|\nabla u\|_{L^2}\|\nabla v\|_{L^2} + C\|\nabla u\|_{L^2}\|v\|_{L^2} \\ &\le M\|u\|_{H_0^1}\|v\|_{H_0^1}\end{aligned}$$

另一方面，
$$\begin{aligned}B[u,u;t] &= \|\nabla u\|_{L^2}^2 + \int_\Omega \sum_{i=1}^d b^i \cdot \partial_{x_i}u \cdot u\,dx \\ &\ge \|u\|_{H_0^1}^2 - C\|u\|_{H_0^1}\|v\|_{L^2} \\ &\ge (1-\tfrac{\varepsilon}{2}C)\|u\|_{H_0^1}^2 - \tfrac{\varepsilon}{2}C\|u\|_{L^2}^2\end{aligned}$$
，取适当 ε 即得结论.

(2) 对方程两边乘 u 并积分得
$$\int_\Omega u u_t\,dx - \int_\Omega u\Delta u\,dx + \sum_{i=1}^d \int_\Omega b^i u\,\partial_{x_i}u\,dx = \int_\Omega f u\,dx$$
由分部积分得，上式即为
$$\frac{1}{2}\frac{\partial}{\partial t}\int_\Omega u^2\,dx + \underbrace{\int_\Omega |\Delta u|^2\,dx + \sum_{i=1}^d \int_\Omega b^i u\,\partial_{x_i}u\,dx}_{B[u,u;t] \overset{(1)}{\ge} \theta\|u\|_{H_0^1}^2 - M\|u\|_{L^2}^2} = \underbrace{\int_\Omega f u\,dx}_{\le \|f\|_{L^2}\|u\|_{L^2} \le \frac{1}{2}\|f\|_{L^2}^2 + \frac{1}{2}\|u\|_{L^2}^2}$$

即 $\frac{\partial}{\partial t}\|u\|_{L^2}^2 \le \|f\|_{L^2}^2 + (1+2M)\|u\|_{L^2}^2$，(*)式

由Gronwall不等式知，对 $\forall t \in [0,T]$，$\|u(t,\cdot)\|_{L^2}^2 \le e^{(1+2M)T}(\|g\|_{L^2}^2 + \|f\|_{L^2(0,T;L^2)})$

故 $\|u(t,\cdot)\|_{L^2} \le \sqrt{2}e^{(\frac{1}{2}+M)T}(\|g\|_{L^2} + \|f\|_{L^2(0,T;L)})$

代回(*)式并做积分即得结论.